Algebra II with Trigonometry Chapter 11.2

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4666322_alg-2trig-h-chp-11-2-note-docx.pdf

Here is a comprehensive study resource based on the provided document.

1. Short Summary

The provided document is an Algebra 2/Trigonometry Honors lesson on ellipses centered at the origin. It introduces the geometric definition of an ellipse as the set of points where the sum of distances to two fixed points (foci) is constant. The text outlines how to identify, graph, and write standard equations for both horizontal and vertical ellipses. It also explores the concept of eccentricity (how "stretched" an ellipse is) and provides various examples requiring students to find equations from specific geometric conditions or solve systems of intersecting ellipses.

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2. Core Definitions, Theorems, and Formulas

Definitions:

- **Ellipse:** An oval curve; geometrically, the set of all points in a plane where the sum of the distances from two fixed points is a constant.
- **Foci (singular: focus):** The two fixed points inside the ellipse used to define its shape.
- **Vertices:** The endpoints of the ellipse along its longest dimension.
- **Major Axis:** The longest segment across the ellipse, joining the two vertices. Its length is always $2a$.
- **Minor Axis:** The shortest segment across the ellipse, perpendicular to the major axis. Its length is always $2b$.

- **Eccentricity (e):** A number between 0 and 1 that measures how "stretched" or elongated the ellipse is. A circle is an ellipse with an eccentricity of 0.

Formulas (Assuming Center is at the Origin and $a > b > 0$):

- **Standard Equation (Horizontal Ellipse):**

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(The major axis is on the x-axis).

- **Standard Equation (Vertical Ellipse):**

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

(The major axis is on the y-axis).

- **Focal Distance Relationship:**

$$c^2 = a^2 - b^2$$

(In words: c^2 = larger denominator minus smaller denominator).

- **Eccentricity:**

$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a}$$

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3. Worked Study Guide: Step-by-Step

As a math tutor, I recommend breaking down ellipse problems into these logical steps:

Step 1: Identify the Orientation (Horizontal vs. Vertical) Look at the standard equation. Find the **larger denominator**. According to the text, the larger denominator is always a^2, and the smaller is b^2.

- If the larger denominator is under x^2, the ellipse is **horizontal**. The vertices and foci will be on the x-axis.
- If the larger denominator is under y^2, the ellipse is **vertical**. The vertices and foci will be on the y-axis.

Step 2: Extract a, b, and c Once you have the equation equal to 1:

- Take the square root of the larger denominator to find a.
- Take the square root of the smaller denominator to find b.
- Use the formula $c^2 = a^2 - b^2$ to find c^2, then take the square root to find c.

Step 3: Map Out the Features Use a, b, and c to locate your points:

- **Vertices** are a units from the center along the major axis.
- The **Minor Axis** endpoints are b units from the center along the other axis.
- **Foci** are c units from the center along the major axis.

Step 4: Understand Eccentricity (e) Calculate $e = \frac{c}{a}$. Remember this is a quick self-check tool!

- Because $c < a$, your answer must be a decimal between 0 and 1.
- If e is close to 1 (e.g., 0.86), the ellipse is long and skinny.
- If e is close to 0 (e.g., 0.1), the ellipse looks almost exactly like a circle.

Step 5: Working Backwards (Given features, find the equation) If you are given conditions (like in Example 3 of the text), treat it like a puzzle.

- Did they give you a vertex at $(0, 5)$? That tells you the ellipse is vertical (because it's on the y-axis) and $a = 5$.
- Did they give you the foci at $(\pm 4, 0)$? That tells you the ellipse is horizontal and $c = 4$.
- Use the relation $a^2 - b^2 = c^2$ to find the missing variable, and construct your equation $\frac{x^2}{?} + \frac{y^2}{?} = 1$.

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4. Practice Problems

These problems are adapted directly from the unworked examples provided in the document.

Problem 1: Find the vertices, foci, and eccentricity of the ellipse given by the equation: $\frac{x^2}{9} + \frac{y^2}{64} = 1$. (Based on Ex 1.i)

- **Answer:**
- The larger denominator is 64, under y^2. So, this is a vertical ellipse.
- $a^2 = 64 \to a = 8$. Vertices are at $(0, \pm 8)$.
- $b^2 = 9 \to b = 3$.
- $c^2 = a^2 - b^2 \to c^2 = 64 - 9 = 55 \to c = \sqrt{55}$. Foci are at $(0, \pm\sqrt{55})$.
- Eccentricity: $e = \frac{c}{a} = \frac{\sqrt{55}}{8}$.

Problem 2: Determine the lengths of the major and minor axes for the ellipse: $3x^2 + 4y^2 = 12$. (Based on Ex 1.ii)

- **Answer:**
- First, divide the entire equation by 12 to set it equal to 1: $\frac{x^2}{4} + \frac{y^2}{3} = 1$.
- $a^2 = 4 \to a = 2$. Length of major axis $= 2a = 4$.
- $b^2 = 3 \to b = \sqrt{3}$. Length of minor axis $= 2b = 2\sqrt{3}$.

Problem 3: Find the standard equation for the ellipse with Foci at $(\pm 4, 0)$ and Vertices at $(\pm 5, 0)$. (Based on Ex 3.a)

- **Answer:**
- The points are on the x-axis, so the ellipse is horizontal.
- Vertices are at $a = 5 \to a^2 = 25$.
- Foci are at $c = 4 \to c^2 = 16$.
- Since $c^2 = a^2 - b^2$, we have $16 = 25 - b^2 \to b^2 = 9$.
- **Equation:** $\frac{x^2}{25} + \frac{y^2}{9} = 1$

Problem 4: Find the standard equation for the ellipse with a length of the major axis of 4, length of the minor axis of 2, and foci on the y-axis. (Based on Ex 3.c)

- **Answer:**

- Foci on the y-axis means it is a vertical ellipse (the larger denominator goes under y^2).
- Major axis $= 2a = 4 \rightarrow a = 2 \rightarrow a^2 = 4$.
- Minor axis $= 2b = 2 \rightarrow b = 1 \rightarrow b^2 = 1$.
- **Equation:** $x^2 + \frac{y^2}{4} = 1$ *(Writing* $\frac{x^2}{1}$ *is also acceptable).*

Problem 5: Find the intersection points of the pair of ellipses: $4x^2 + y^2 = 4$ $4x^2 + 9y^2 = 36$ (Based on Ex 4.a)

- **Answer:**
- Use elimination. Subtract the first equation from the second:

$(4x^2 + 9y^2) - (4x^2 + y^2) = 36 - 4$ $8y^2 = 32$ $y^2 = 4 \rightarrow y = \pm 2$.

- Substitute $y^2 = 4$ back into the first equation:

$4x^2 + (4) = 4$ $4x^2 = 0 \rightarrow x = 0$.

- **Points of intersection:** $(0, 2)$ **and** $(0, -2)$.

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5. Assumptions and Resolved Ambiguities

To provide accurate information based strictly on this document, I made the following resolutions:

- **Notation Convention (a vs b):** Different textbooks handle vertical ellipses differently. This specific document dictates that a **is always strictly greater than** b, and that a^2 is always the *larger denominator*. When the ellipse is vertical, the document swaps the denominators, putting a^2 under y^2 ($\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$). I used this exact convention in the study guide and practice problems.
- **Example 2 Graphs:** I omitted Example 2 (finding equations from images of graphs) from the practice problems because the exact points/intercepts can be visually ambiguous without grid lines, relying instead on the clearly stated algebraic conditions in Example 3.
- **Image Legibility:** In Example 3e, the point is written with a specific square root format. I assumed visual reading of $(\sqrt{5}, \sqrt{40})$ based on

standard math typography, resolving any minor OCR errors in the provided text extraction.